THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL UNIT TEST SET 3

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The sum of the series $11 + 17 + 23 + \cdots + 113$ is: 1

(a) 1118 **(b)** 1116 **(c)** 1117 **(d)** 1115

2. In $\triangle ABC$ right-angled at B, $AB = 5$, $BC = 12$. If $BD \perp AC$, then AD (cm, correct to the exact value) is: 1

(a) $\frac{51}{13}$ **(b)** $\frac{38}{13}$ **(c)** $\frac{12}{13}$ **(d)** $\frac{25}{13}$

3. The side of the square with diagonal endpoints $(1, 1)$ and $(7, 9)$ has squared length: 1

(a) 51 **(b)** 49 **(c)** 50 **(d)** 52

4. How many terms of the AP $18, 15, 12, 9, \ldots$ give sum 0? 1

(a) 13 **(b)** 12 **(c)** 15 **(d)** 14

5. A square has diagonal $7\sqrt{2}$ cm. Its area (cm²) is: 1

(a) 51 **(b)** 49 **(c)** 48 **(d)** 50

6. The three points $(1, 1)$, $(9, 1)$ and $(4, 8)$ enclose area: 1

(a) 27 **(b)** 29 **(c)** 28 **(d)** 30

7. If the 3rd term of an AP is 8 and the 8th term is 3, then the 11th term is: 1

(a) 24 **(b)** 0 **(c)** 1 **(d)** 11

8. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 3 : 4$. The ratio of areas of $\triangle ADE$ to trapezium $DBCE$ is 9: 1

(a) 39 **(b)** 42 **(c)** 40 **(d)** 41

9. The point on the y-axis equidistant from $(3, 1)$ and $(7, 7)$ has $y =$ 1

(a) $\frac{19}{3}$ **(b)** $\frac{28}{3}$ **(c)** $\frac{22}{3}$ **(d)** $\frac{25}{3}$

10. The sum of the first 15 terms of an AP is 465, and its first term is 10. The 15th term is: 1

(a) 54 **(b)** 53 **(c)** 51 **(d)** 52

11. In a right triangle with hypotenuse $c = 17$ and one leg $a = 8$, the area (sq. units) is: 1

(a) 60 **(b)** 62 **(c)** 61 **(d)** 59

12. The area of the triangle formed by $(6, 0)$, $(0, 13)$ and the origin, doubled, is: 1

(a) 80 **(b)** 77 **(c)** 78 **(d)** 79

13. In an AP, $S_{12} = 564$ and $S_{11} = 473$. Then $a_{12} =$ 1

(a) 91 **(b)** 90 **(c)** 92 **(d)** 93

14. The altitude on the hypotenuse of the right triangle with legs 8 and 15 is of length (cm): 1

(a) $\frac{137}{17}$ **(b)** $\frac{103}{17}$ **(c)** $\frac{154}{17}$ **(d)** $\frac{120}{17}$

15. If $P(x, y)$ is equidistant from $(0, 6)$ and $(10, 2)$ and lies on $y = 0$, then $x =$ 1

(a) $\frac{22}{5}$ **(b)** $\frac{17}{5}$ **(c)** $\frac{27}{5}$ **(d)** $\frac{12}{5}$

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. In $\triangle ABC$, $DE \parallel BC$, $AD = 2$ cm, $AB = 7$ cm and $BC = 21$ cm. Find DE. 2

17. Show that the points $A(1, 2)$, $B(4, 6)$, $C(9, 6)$, $D(6, 2)$ are the vertices of a rhombus. 2

18. If $S_n = 2n^2 + 5n$, find the first term and the common difference. 2

19. The areas of two similar triangles are 32 cm² and 162 cm². If a side of the first is 4 cm, find the corresponding side of the second. 2

20. Find the value of k for which the points $(-2, 3)$, $(0, 7)$ and $(3, k)$ are collinear. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. The line joining $(3, 3)$ and $(12, 9)$ is trisected by points P and Q. Find their coordinates. 3

22. Find the sum of all integers between 20 and 300 that leave remainder 4 when divided by 9. 3

23. $\triangle ABC$ is right-angled at B with $AB = 5$ cm and $BC = 12$ cm; $BD \perp AC$. Find AD, DC and BD. 3

24. Find the values of y for which the distance between $P(17, y)$ and $Q(5, 1)$ is 13 units. 3

OR

In an AP, $S_7 = 98$ and $S_{12} = 228$. Find the AP. 3

25. In trapezium $ABCD$, $AB \parallel DC$ and the diagonals meet at O. If $AB = 5$ cm and $DC = 20$ cm, find $AO : OC$ and the ratio $ar(AOB) : ar(COD)$. 3

OR

Show that the points $A(2, 3)$, $B(5, 7)$, $C(10, 7)$, $D(7, 3)$ are the vertices of a rhombus. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. The vertices of $\triangle ABC$ are $A(1, 2)$, $B(7, 2)$ and $C(4, 8)$. Find the coordinates of the centroid and the area of the triangle. 5

OR

A loan is repaid in 40 monthly instalments; the first is ₹600 and each later instalment is ₹45 more than the previous. Find the instalment in the last month and the total amount repaid. 5

27. Two poles of heights 45 m and 9 m stand on level ground 27 m apart. Find the distance between their tops. 5

OR

Three vertices of a parallelogram $ABCD$ taken in order are $A(1, 5)$, $B(7, 3)$ and $C(10, 9)$. Find the fourth vertex D. 5

END OF QUESTION PAPER — **Total: 50 Marks**